COMP3308/3608 Introduction to Artificial Intelligenece
(regular and advanced)

semester 1, 2023

Information about the exam

- There are two types of exams: in-person and online, depending on your enrolment
  - CC students will have an in-person, paper-based exam
  - RE students will have an online exam
- Both exams are supervised - by exam supervisors in the room (in-person exams) or remotely via the ProctorU's Live+ system (online exams)
- Permitted materials for both exams:
  - One sheet of your own notes - double-sided A4 size, typed or handwritten - paper-based, not electronic
  - Calculator - handheld, non-programmable
  - In addition, for online exams only: blank scratch paper, multiple sheets. This is not permitted for paper-based exams - blank pages are already included at the end of the exam paper - use them if needed.
- No other materials or devices are allowed. You are not allowed to use your phone during the exam. You are not allowed to browse the web, access files or programs on your computer. You can't consult other people during the exam in any way.
- For both exams, you need to have a photo identification - Student ID card or other accepted documents
- In-person exam:
  - There are specific rules about the in-person exams, read them carefully:
    https://www.sydney.edu.au/students/exams/in-person.html
  - The exam paper is like a booklet. You write your answers on the exam paper, in the space provided. There is a question, then space for you to write the answer; another question and space for the answer, etc. To write the answers you should use black or blue pen, not pencil.
- Online exam:
  - It is set in Canvas as a Quiz. The Canvas site for the exam is different that the Canvas site we use during the semester. The Exams Office will give you access to the exam site.
  - It is type Live+. Please read “Preparing for online exams”, select Live+:
    https://canvas.sydney.edu.au/courses/23380
  - Help and FAQs about Live+ exams:
    https://canvas.sydney.edu.au/courses/23380/pages/live+-help-and-faqs
  - You need to type your answers in the boxes in the Quiz or select the correct answer from a list for the multiple choice questions. You are not allowed to upload images. This means that you are not allowed to scan a handwritten solution with your phone and attach it.

- The exam paper is **confidential**. You must not discuss the exam questions with other people, post or distribute the exam questions in any way, both during or after the exam.
- The duration of the exam is standard: 2 hours + 10 minutes reading time.
- The exam is worth 60 marks (=60% of your final mark). To pass the course you need at least 40% on the exam (i.e. 24 marks), regardless of what your mark during the semester is.
- All material is examinable except week 1, week 13a (recommender systems), historical context, Matlab and Weka.
- There are 3 types of questions: 1) questions requiring short answers, 2) problem-solving/calculation questions and 3) multiple-choice questions (small number).

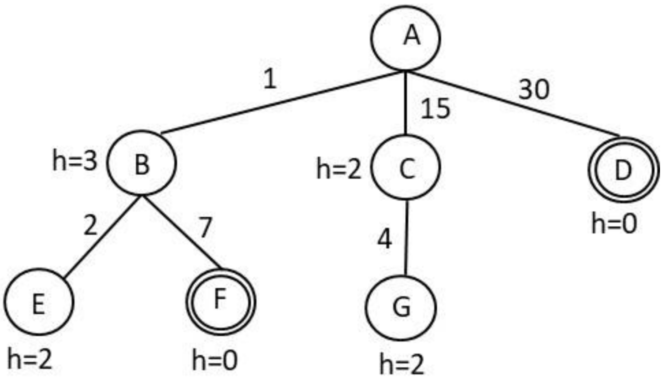
Academic honesty - Important

- Both exams, in-person and online, are supervised
- All suspicious behavior during the online exam will be reviewed
- **Please do not cheat or copy!** The consequences and penalties are very severe.
- If you copy, you will get caught. If you make your work available to another student to copy, this is also academic dishonesty and you will be investigated and penalized.
- The stress of going through the investigation is immense. The investigation takes many months and your mark will not be finalised until it is completed, your graduation may be delayed, you may have problems enrolling in other courses.
- **It is not worth it. You will regret it all your life.**

Sample exam questions

In addition to these questions please also see on Canvas:
Search: Weeks_2-3_Practice.pdf (prepared by Jessica)
Bayesian networks: BN_practice_questions.pdf (prepared by Jessica)

Question 1. (Type 2 – problem solving/calculation)
In the tree below the step costs are shown along the edges and the h values are shown next to each node. The goal nodes are double-circled: F and D.



Write down the order in which nodes are expanded using:

a) Breadth-first search
b) Depth-first search
c) Uniform cost search
d) Iterative deepening search
e) Greedy search
f) A*

In case of ties, expand the nodes in alphabetical order.

Question 2. (Type 1 – short answers)

Answer briefly and concisely:

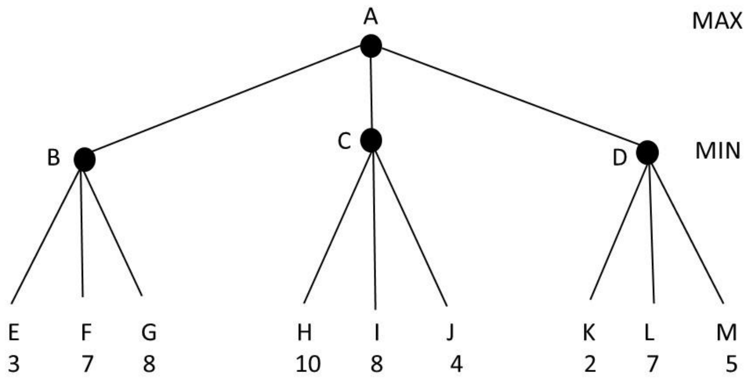
a) A* uses admissible heuristics. What happens if we use a non-admissible one? Is it still useful to use A* with a non-admissible heuristic?

b) What is the advantage of choosing a dominant heuristic in A* search?

c) What is the main advantage of hill climbing search over A* search?

Question 3. (Type 2 – problem solving/calculation)

Consider the following game in which the evaluation function values for player MAX are shown at the leaf nodes. MAX is the maximizing player and MIN is the minimizing player. The first player is MAX.



a) What will be the backed-up value of the root node computed by the minimax algorithm?

b) Which move should MAX choose based on the minimax algorithm – to node B, C or D?

c) Assume that we now use the alpha-beta algorithm. List all branches that will be pruned, e.g. AB etc. Assume that the children are visited left-to-right (as usual).

Question 4. (Type 1 – short answers)

Answer briefly and concisely:

a) The 1R algorithm generates a set of rules. What do these rules test?

b) *Gain ratio* is a modification of *Gain* used in decision trees. What is its advantage?

c) Propose two strategies for dealing with missing attribute values in learning algorithms.

d) Why do we need to normalize the attribute values in the k-nearest-neighbor algorithm?

e) What is the main limitation of the perceptrons?

f) Describe an early stopping method used in the backpropagation algorithm to prevent overfitting.

g) The problem of finding a decision boundary in support vector machine can be formulated as an optimisation problem using Lagrange multipliers. What are we maximizing?

h) In linear support vector machines, we use dot products both during training and during classification of a new example. What vectors are these products of?

During training:

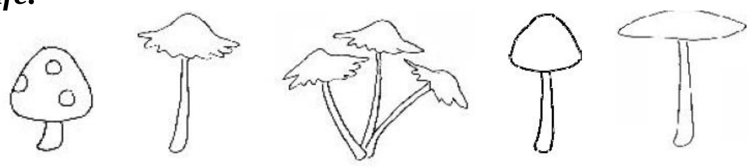
During classification of a new example:

Question 5. (Type 2 – problem solving/calculation)

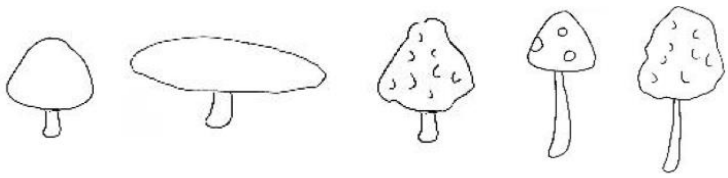
Consider the task of learning to classify mushrooms as ***safe*** or ***poisonous*** based on the following four features: stem = {short, long}, bell = {rounded, flat}, texture = {plain, spots, bumpy, ruffles} and number = {single, multiple}.

The training data consists of the following 10 examples:

Safe:



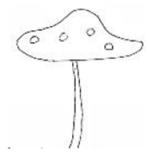
Poisonous:



These examples are also shown in the table below:

n	stem	bell	texture	number	class
1	short	rounded	spots	single	safe
2	long	flat	ruffles	single	safe
3	long	flat	ruffles	multiple	safe
4	long	rounded	plain	single	safe
5	long	flat	plain	single	safe
6	short	rounded	plain	single	poisonous
7	short	flat	plain	single	poisonous
8	short	rounded	bumpy	single	poisonous
9	long	rounded	spots	single	poisonous
10	long	rounded	bumpy	single	poisonous

a) Use Naïve Bayes to predict the class of the following new example: stem=long, bell=flat, texture=spots, number=single. Show your calculations.



a) How would 3-Nearest Neighbor using the Hamming distance classify the same example as above? Explain your answer. (Hint: The Hamming distance is the number of different feature values).

b) Consider building a decision tree. Calculate the information gain for ***texture*** and ***number*** and briefly show your calculations Which one of these two features will be selected and why?

You may use this table:

x	y	-(x/y)*log2(x/y)	x	y	-(x/y)*log2(x/y	x	y	-(x/y)*log2(x/y	x	y	-(x/y)*log2(x/y
1	2	0.50	4	5	0.26	6	7	0.19	5	9	0.47
1	3	0.53	1	6	0.43	1	8	0.38	7	9	0.28
2	3	0.39	5	6	0.22	3	8	0.53	8	9	0.15
1	4	0.5	1	7	0.40	5	8	0.42	1	10	0.33
3	4	0.31	2	7	0.52	7	8	0.17	3	10	0.52
1	5	0.46	3	7	0.52	1	9	0.35	7	10	0.36
2	5	0.53	4	7	0.46	2	9	0.48	9	10	0.14
3	5	0.44	5	7	0.35	4	9	0.52			

d) Consider a single perceptron for this task. What is the number of inputs? What is the dimensionality of the weight space? Briefly explain your answer.

Question 6. Given the training data in the table below where *credit history*, *debt*, *collateral* and *income* are attributes and *risk* is the class, predict the class of the following new example using the 1R algorithm: *credit history=unknown, debt=low, collateral=none, income=15-35K.* Show your calculations.

credit history	debt	collateral	income	risk
bad	high	none	0-15k	high
unknown	high	none	15-35k	high
unknown	low	none	15-35k	moderate
unknown	low	none	0-15k	high
unknown	low	none	over 35k	low
unknown	low	adequate	over 35k	low
bad	low	none	0-15k	high
bad	low	adequate	over 35k	moderate
good	low	none	over 35k	low
good	high	adequate	over 35k	low
good	high	none	0-15k	high
good	high	none	15-35k	moderate
good	high	none	over 35k	low
bad	high	none	15-35k	high

Question 7. (Type 2 – problem solving/calculation)

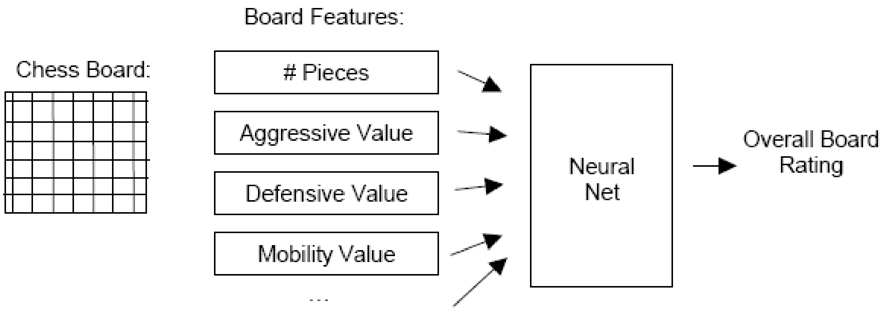
Use the k-means algorithm to cluster the following one dimensional examples into 2 clusters: 2, 5, 10, 12, 3, 20, 31, 11, 24. Suppose that the initial seeds are 2 and 5. The convergence criterion is met when either there is no change between the clusters in two successive epochs or when the number of epochs has reached 5.

Show the final clusters. How many epochs were needed for convergence? There is no need to show your calculations.

Question 8. (Type 1 – short answers)

You task is to develop a computer program to rate chess board positions. You got an expert chess player to rate 100 different chessboards and then use this data to train a backpropagation neural network, using board features as the ones shown in the figure below.

Select the correct answer (“Yes” or “No”) in the questions below. Select “Yes” for all issues that could, in principle, limit your ability to develop the best possible chess program using this method. Select “No” for all issues that could not. Briefly explain your answer.



a) The backpropagation network may be susceptible to overfitting, since you tested its performance on the training data instead of using cross validation.

Yes No

Explanation:

b) The backpropagation neural network can only distinguish between boards that are completely good or completely bad.

Yes No

Explanation:

c) The backpropagation network will converge to the global minimum.

Yes No

Explanation:

d) You should have used higher learning rate and momentum to guarantee convergence to the global minimum.

Yes No

Explanation:

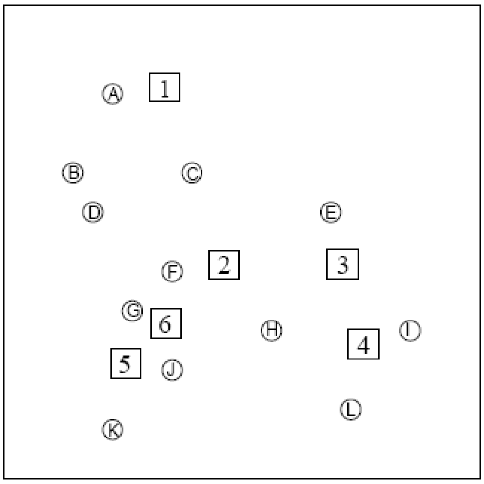
e) The topology of your neural net might not be adequate to capture the expertise of the human expert.

Yes No

Explanation:

Question 9. (Type 2 – problem solving/calculation)

In the figure below, the circles are training examples and the squares are test examples, i.e. we are using the circles to predict the squares. Two algorithms are used: 1-Nearest Neighbour and 3-Nearest Neighbour.



We are given the following results about the squares:

Square	Using 1-Nearest Neighbors	Using 3-Nearest Neighbors
1	-	+
2	-	
3		+
4	+	-
5		-

What will be the class of the following examples? Write +, - or U for cannot be determined.

1) Circle L: -
2) Circle I: +
3) Circle H: -
4) Circle E: +
5) Circle K: U
6) Circle C: +
7) Square 6 using 1-Nearest Neighbour: U
8) Square 6 using 3-Nearest Neighbour: -
9) Square 3 using 1-Nearest Neighbour? +
10) Square 5 using 1-Nearest Neighbour? U